Installation Manual

Vertical lifting fabric door
Crawford VL3016

ASSA ABLOY

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door opening solutions



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1. Introduction

1.1 General information

The installation instructions for Crawford VL3016 vertical lifting fabric door are compiled in conformance with European standard EN 12635:2002 + A1:2008 “Industrial, commercial and garage doors – Installation and use”.
The door and its components shall be fitted in compliance with these installation instructions.

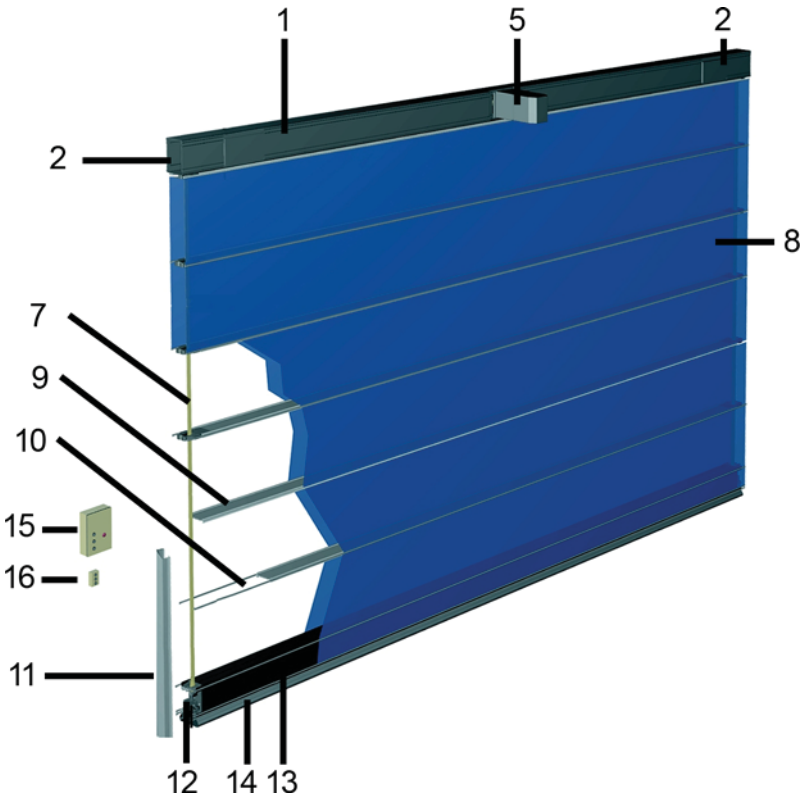
Pay special attention what is written about mounting under the headings “Installation - general”, “Guide rails - mounting surfaces” and “Header box - mounting surfaces”.

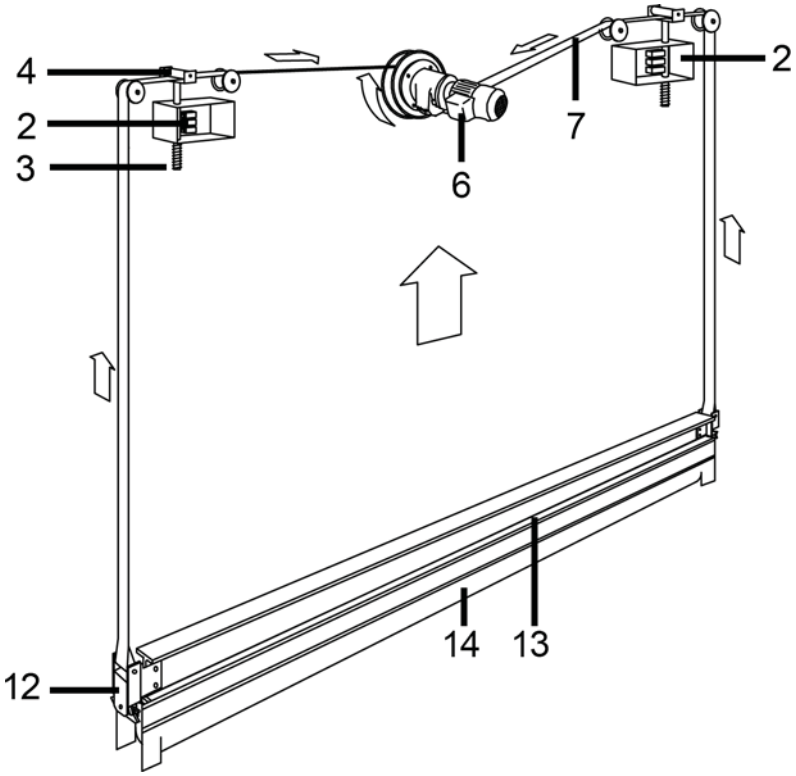
Instructions according to EN 12604:2000 “Mechanical aspects – requirements” and EN 12453:2000 “Safety in use of power operated doors - requirements” must be followed and verified if necessary.

The door is intended for installation by a professional installer. Installation carried out by anyone other than professional installers could result in a non conforming door and also risk of personal injury.

Personnel who have limited training may assist the installer.

1.2 Door leaf and header box parts list





1) Header box
2) Limit switch box
3) Push rod
4) Pull rod
5) Gear motor
6) Brake lever
7) Lifting belt
8) Door leaf
9) Intermediate section
10) Clamp strip
11) Guide rail
12) Safety arrester**
13) Bottom section
14) Floor seal
15) Control cabinet
16) Push button unit

** The safety arrester, made of stainless steel, is activated during each opening sequence and also locks the door in the closed position.

1.3 Delivery

A Crawford VL3016 vertical lifting fabric door is usually delivered in two packages:

1) Guide rail / jambs
2) Door leaf and bottom section
3) Header box with drive unit (1 or 2 pcs), motor casing (if any) and fixing parts.Control cabinet, electrically connected and test run.





2. Installation

2.1 General

1) The door is installed in the building by screw fitting.
2) Screws are to be torque tightened according to the table below (± 5%):



Do not use impact tools

Thread forming screw, taptite:

Thread	Drill diameter	Material thickness	Tightening torque
M5	4.5 mm	1.5 -2.5 mm	6.5 Nm
M6	5.5 mm	2.5 - 4.0 mm	11.0 Nm
M8	7.4 mm	4.0 - 6.5 mm	26.5 Nm
M10	9.3 mm	6.5 - 10.0 mm	52.5 Nm

Conventional screw (class 8):

Thread	Drill diameter	Tightening torque
M8	6.8 mm	24 Nm
M10	8.5 mm	47 Nm
M12	10.2 mm	81 Nm

3) Electrical installations are to be made professionally and in conformance with national requirements.
4) Ensure that the door opening conforms to the installation documents that accompany each order and also that the dimensional tolerances stated there, stability and other directions are followed accordingly.

Tools
Recommended installation tools.

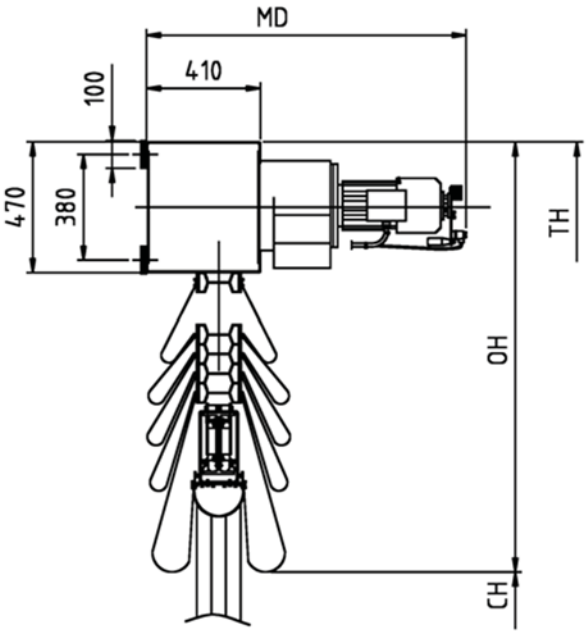
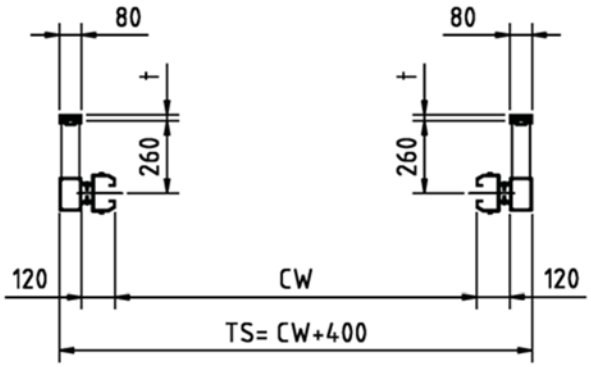
Suitable lifting equipment
To lift personnel: articulated lift, skylift or scaffolding.
To lift material: forklift truck, wheel loader or small crane.

2.2 Installation options

The Crawford VL3016 vertical lifting fabric door can be installed in accordance with the following options. See installation documents for the delivery.

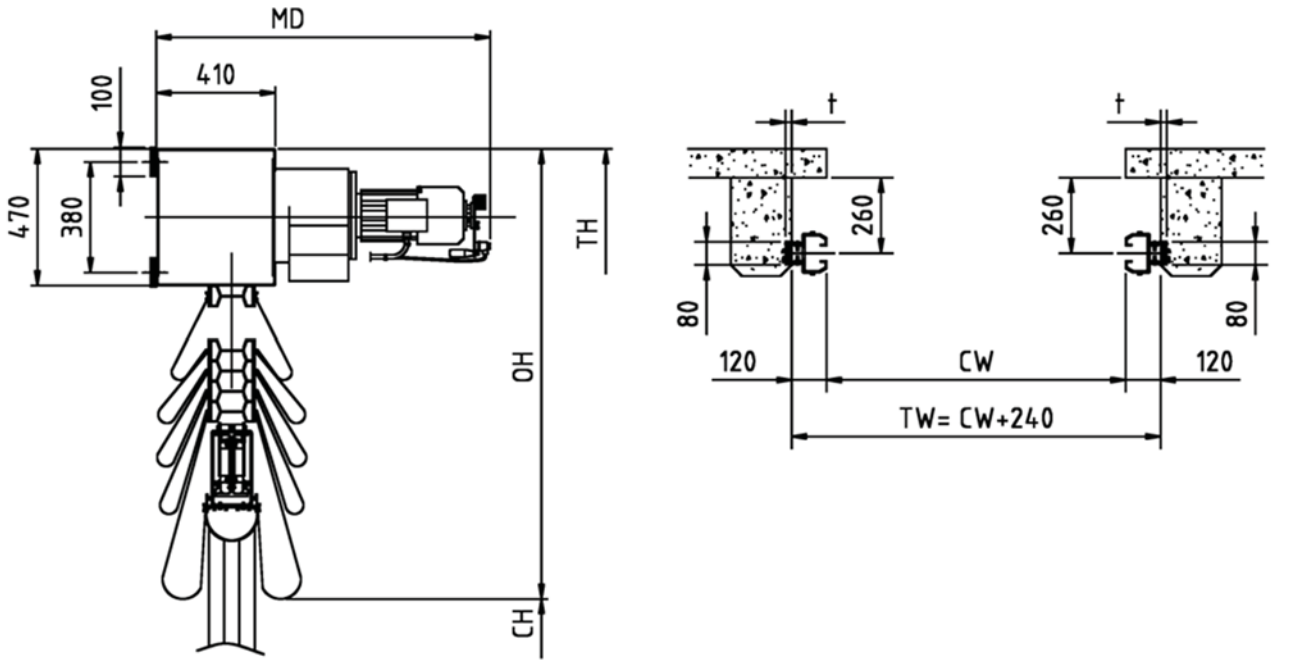
Abbreviation explanation:

TH:	Total height	TW:	Total width (± 10 mm)
OH:	Over height	CW:	Clear width
CH:	Clear height	MD:	Motor depth
TS:	Total space requirement (± 10 mm)	T:	Thickness acc. to "Fastening"

Header box	Guide rails
	

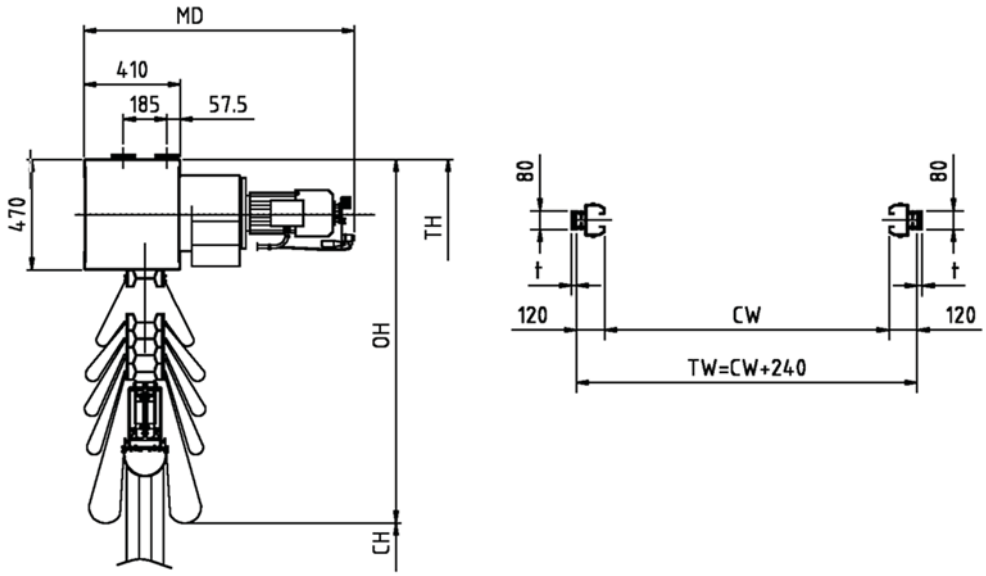
Installation against wall with jambs

Header box **Guide rails**



Installation against wall between columns

Header box **Guide rails**



Installation in door opening

2.3 Guide rails - mounting surfaces

Suitable mounting surfaces must be available to facilitate the fitting of guide rails (see dark fields in the illustration). Mounting surfaces must be:

- Firm and smooth.
- Parallel and deviating not more than 5 mm from the vertical and not more than 2 mm/m in the inward or outward direction from the vertical.

Distance between fixing points should not exceed 1m.

Check for any slope of the floor to determine at which level the fitting of the guide rails is to begin.

The difference in height may not exceed 1 mm/m of the door width.

Fastening:

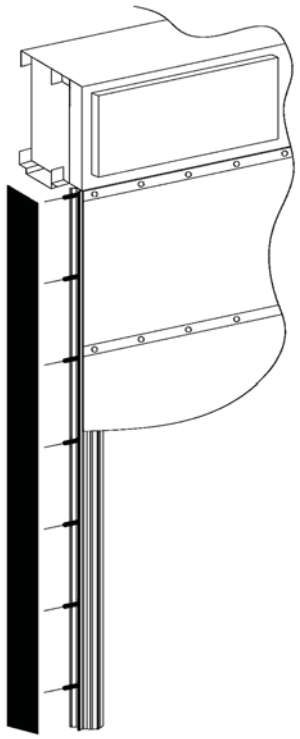
Screw size: M10

Strength class: 8.8

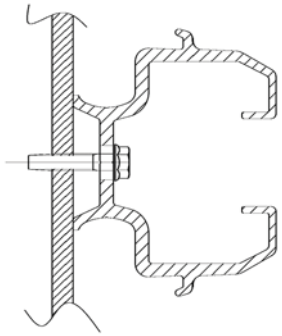
Fixing surface, steel: t ≥ 8

Fixing surface, concrete: t ≥ 100

See also "Installation options"

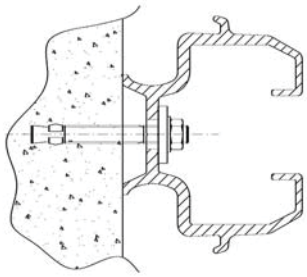


Steel:



Drill holes to thread with M10 tap or thread froming screw M10.

Concrete:



Drill holes for M10 expansion screw.

Do not exceed the recommended tightening torque (applies to both steel and concrete).

Do not use impact tools.

2.4 Guide rails - mounting



1) Mark a vertical line for the fixing points.



2) Drill holes for the screws. If the floor slopes, begin at the side that is highest. Then drill screw holes on the other side at the same levels.



3) Begin by fitting the shortest guide rail. Leave about 2 mm between the rail and the floor.

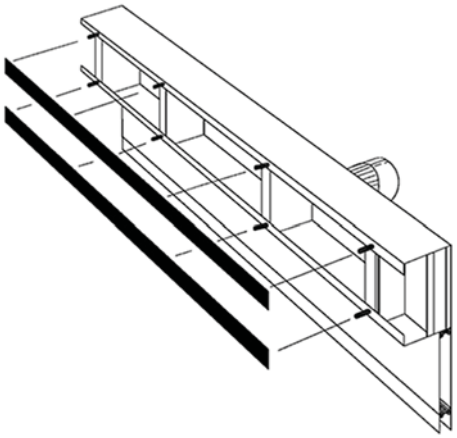


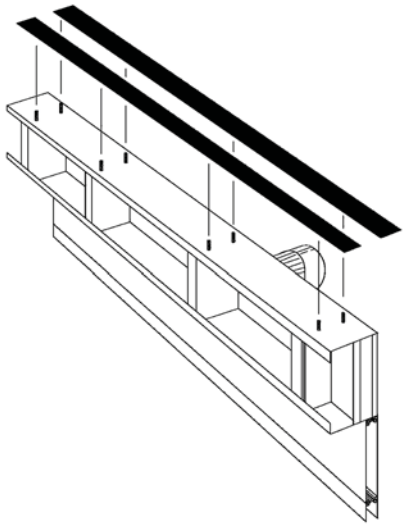
4) Ensure that the guide rails are vertical.



2.5 Header box - mounting surfaces

The rear or top of the header box may be secured to the wall or beams of the building by using screws, welding or beam clamps.

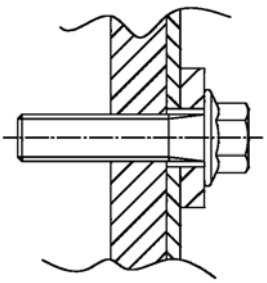




Mounting on wall
Flat, vertical or horizontal surfaces must be available to facilitate fitting the back of the header box (see dark fields in the illustration).

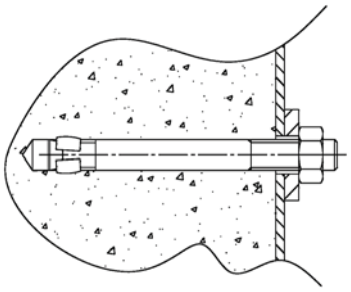
Mounting in opening
Flat, horizontal surfaces must be available to facilitate fitting at the top of the header box (see dark fields in the illustration).

Steel:



Drill holes for thread forming screw M10 or conventional M12 to secure the header box.

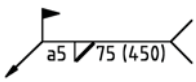
Concrete:



Drill holes for expansion screw M12 to secure the header box.

Fastening:
Screw size: M12 or M10 taptite
Strength class: 8.8
Fixing surface, steel: t ≥ 10
Fixing surface, concrete: t ≥ 110
See also "Installation options"

Hole pattern header box fastening: If information is required, please contact Crawford

Welding: 

a5x250 at both ends

2.6 Header box - mounting



1) Dismantle the front cladding of the header box. Lift up the header box, it is important that the entire length is supported.

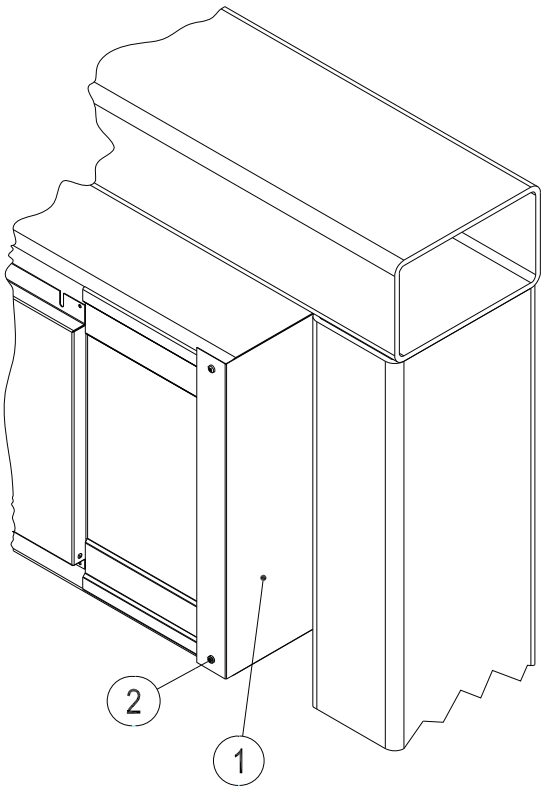


2) Place the header box on top of the already installed guide rails; centered in relation to the opening. Ensure that the header box is levelled. The difference in height between the ends may not exceed 1 mm per metre width of the header box.
If the header box shall be fastened with screws, there are predrilled holes in the header box (rear and top). Seal the holes that not are used with the enclosed plastic stoppers. If beam clamps are used for fastening (top mounting), the maximum distance is 1000 mm.
For welding, use the dimensions 5x75 (450) mm, at the ends of the header box 5x250 mm.





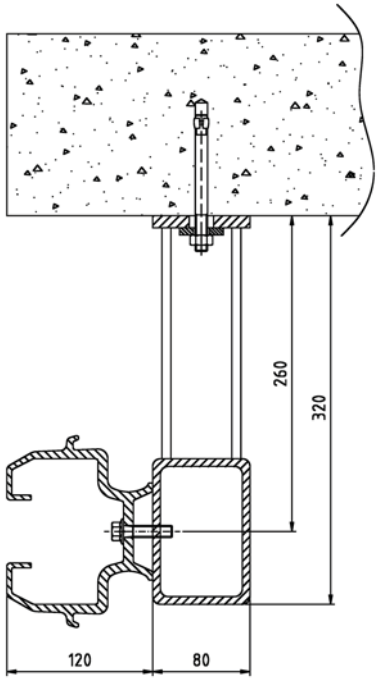
3) Lift up the motor and screw it to the header box. Screw the 2 limit switch units. Loosen and feed the belt down through the holes in the header box and on down in the guide rails.



4) If necessary, trim supplied cover plates (1) to fit, and cover the header box gables. Use self tapping screw M5x16 (2), also included in package, to fasten them.

2.7 Examples of available options

- Protective motor casing. In outdoor installation or unclean environment.
- Enclosed “non-motor side”. For installation in opening with the motor facing inwards, the open rear of the header box (facing outwards) must be provided with fixed cladding sheets.
- Jambs – see Installation options. Used when there are no fixing surfaces for the guide rails.



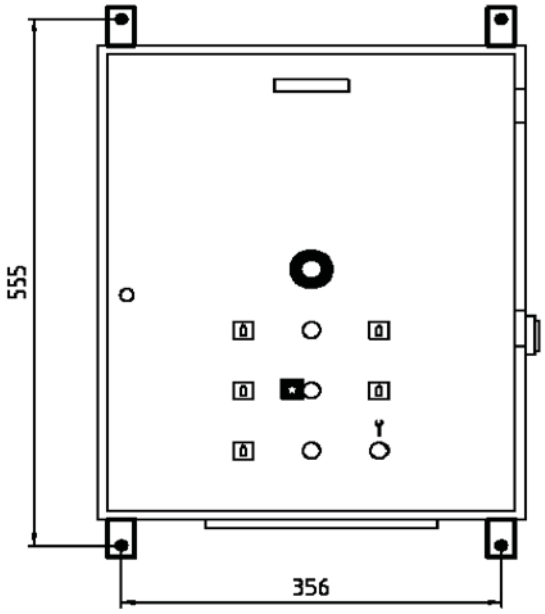
Jamb

- Vision panels of different sizes, different types of special fabric.
- Increased opening speed *.
- Automatic closing function with photocells, magnetic loops, etc *.
- Radio control functions *.
- Etc.

* Instructions and functional descriptions accompany the respective function option

2.8 Control cabinet and cable





1) Release the control cabinet from the header box and place it in position according to the documentation. Permanent power supply is required to complete the installation according to EU regulations
Ensure that the mains fuses are not higher rated than what is approved for the circuit breaker.

2) Clamp the cables from the control cabinet and upward. Distance between clamps about 25 cm; from each bend about 10 cm. Lay cable only vertically and horizontally.

Do not hammer the cables.
Roll up any remaining cable inside the header box.





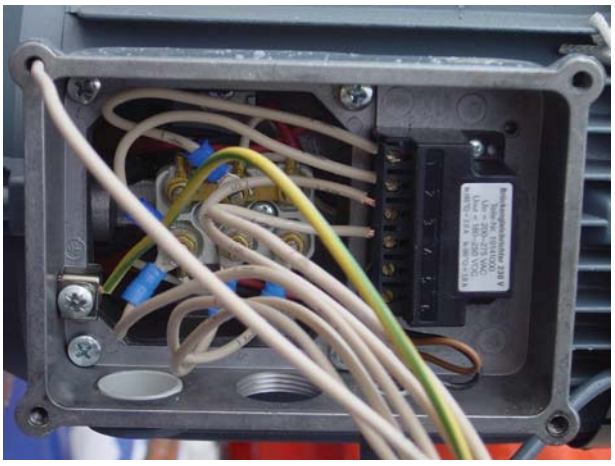
Crank interlock

3) For additional push button station, use protective conduit through sheet-metal wall or other light wall.

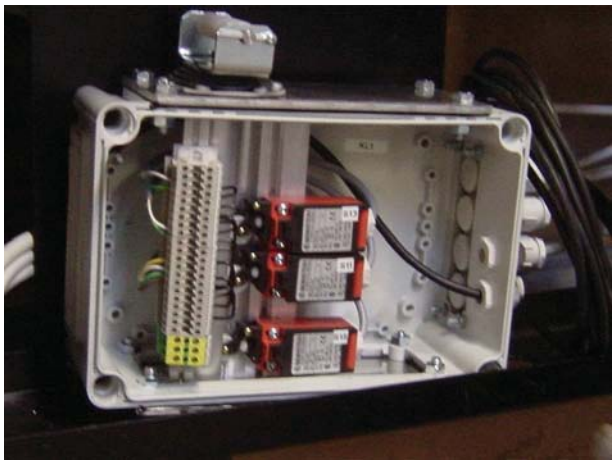


4) The cables are normally preinstalled at the factory. If this is not the case, a 7-wire cable shall be installed from the control cabinet to the gear motor (2 cables if 2 motors) and a 10-wire cable to the left limit switch box. The left limit switch box has terminal blocks, right has not. Three 3-wire cables are installed between left and right limit switch box. A 2-wire cable is installed from the crank interlock to the left limit switch box. Connect all the cables according to the wiring diagram.

At start-up, all terminal block connections must be retightened to avoid bad contact!



Gear motor connection



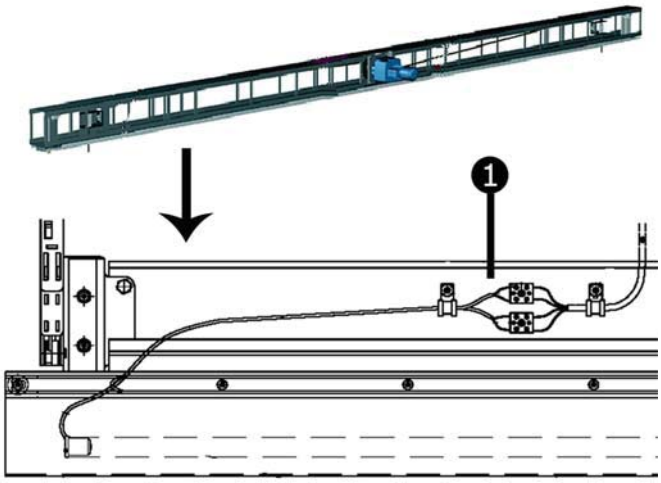
Left limit switch box (KL1)

After any change to the mains power supply, ensure that the phase order is correct. Run the door with the "Down" push-button. If the door goes up, stop it immediately by pushing the Stop or Emergency Stop push-button. Turn off the power supply to the door and then change two phases in the main switch or, if not existing, on the terminal block in the control cabinet of the door.

2.9 Bottom section



1) Remove the short guide rail

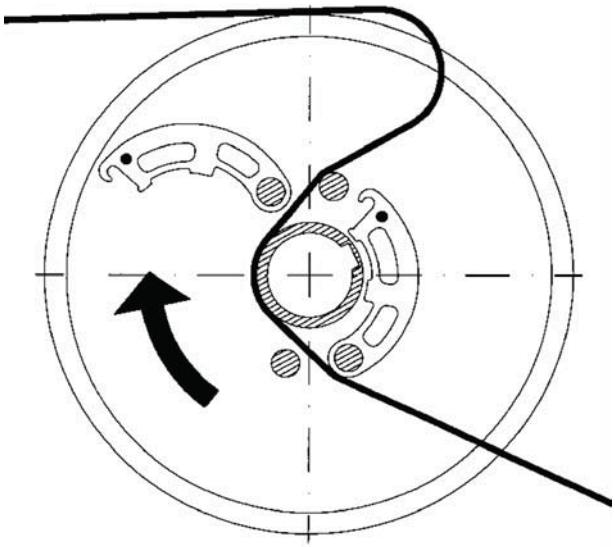


1) Terminal blocks

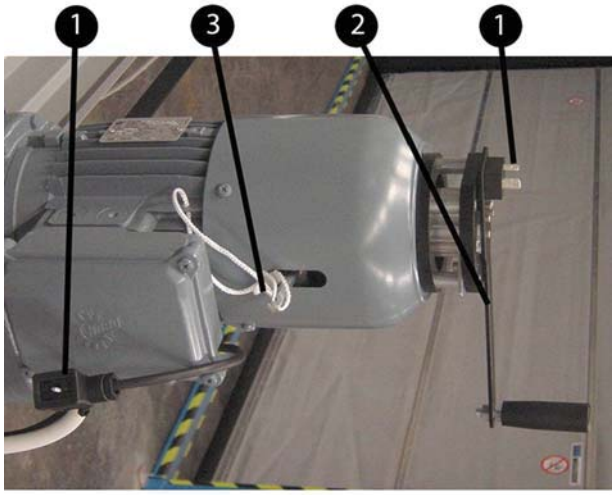
2) Lift in the bottom section, ensuring that the 4 terminal blocks for the safety-edge cable is to the left of the motor on the motor side



3) Secure the lifting belts in the safety arresters, ensuring that the retainer ring enters its groove in the shaft. The opening of the retainer ring shall face down.



4) Ensure that the bottom section is horizontal. If not, adjust the belt length by cranking the motor to release the belt segment as illustrated.



5) Disconnect the crank interlock (1) on the hand crank (2). Detach the crank. Turn the crank and mount it on the shaft. Attach the brake lever (3). Firmly grip the crank. Cautiously release the brake by means of lever and start cranking.



6) Measure the difference between both ends of the bottom section. Displace the belt half the measured difference towards the higher side of the bottom section.



7) Ensure that the bottom section is now horizontal and sealing against the floor. Fix the belt by turning the belt drum in the direction of the arrow (see illustration #4) so that the belt is wound up and locked by the segments.

For test, run the bottom section up into the guide rail. The bottom section shall run smoothly without any noise or tendency to stick. Lower the bottom section to floor level after finishing the test and slacken the belt.

2.10 Door leaf



1) Fit the mounting guides (installation equipment) at the far bottom under the long guide rails.



2) Lift in the door leaf with packing (and pallet if any) and place it on the bottom section. If the door leaf is heavy, attach two straps round it so that it can be lifted into the door opening using a fork lift or by other means.



3) Ensure that the door leaf is facing the right way so that the top section screws will fit into the holes in the header box.



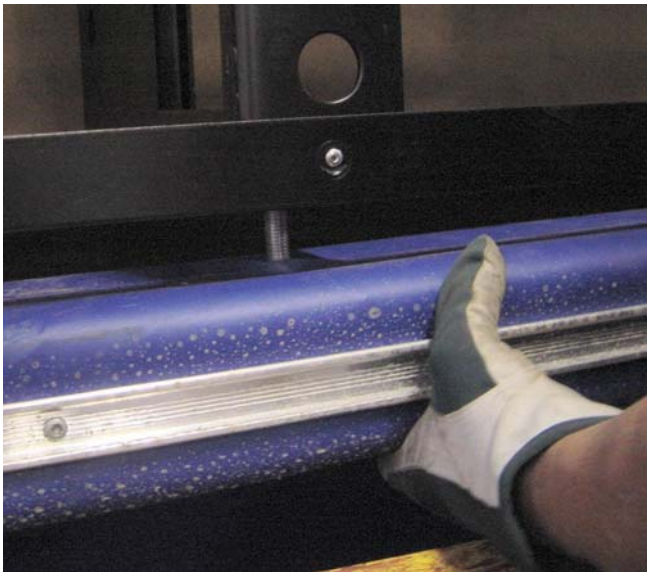
4) Lift up the door leaf and guide it carefully into the guide rails. Lift it enough to make room for the bottom section beneath it. Remove the screws, holding the transport securing devices.



5) Make certain that the fabric, clamp strips and nylon knobs are not damaged.



6) Run the bottom section up to the door leaf. Before it reaches the door leaf, place out blocks on the bottom section where suitable, to create space so that the straps can be removed. When the door leaf is resting on the bottom section, remove the straps and any remaining packing and remove the transport securing devices.



7) Carefully run the door leaf up and stop 200 mm from the bottom of the header box. Crank the motor to lift the door leaf and centre the top section in relation to the header box so that the screws and the push rods get into position.



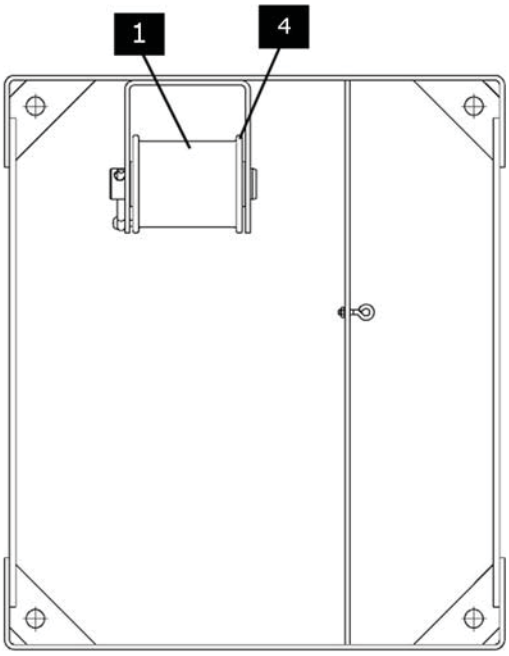
8) Screw the door leaf firmly to the header box and secure the short guide rails.

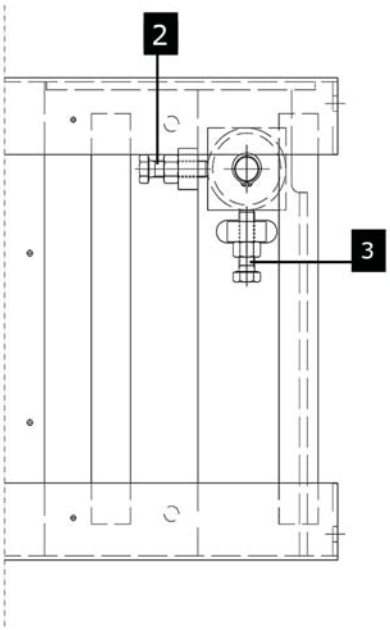
Do not exceed the prescribed tightening torque. Remove the mounting guides and secure the short guide rails.

2.11 Rollers

The two belt rollers (1) are not factory adjusted and must be adjusted during the installation. A preliminary adjustment should be done while test running the bottom profile before the door installation is completed. Before starting the adjustment, check that the header box is mounted straight both horizontally and vertically. A badly installed or crooked header box can make adjustment impossible. The adjustment mechanism in both ends of the header box has two adjustment screws, one horizontal (2) and one vertical (3). The horizontal screw is used for adjusting the tracking of the belt during closing. The vertical screw is used for adjusting the tracking of the belt during opening. Tighten a screw will move the belt towards the rear side of the header box. Untighten a screw will do the opposite. In order to optimize the tracking of the belt, several adjustments may be necessary, as the belt might not be running in the centre of the roller during the whole cycle. When adjustment is completed, tighten the nut to secure the setting. After door leaf being installed, a final adjustment must be done.
A correct adjustment means that:

- The belt runs centred on the roller.
- The runs under no circumstances start climbing the ridge (4) of the roller.

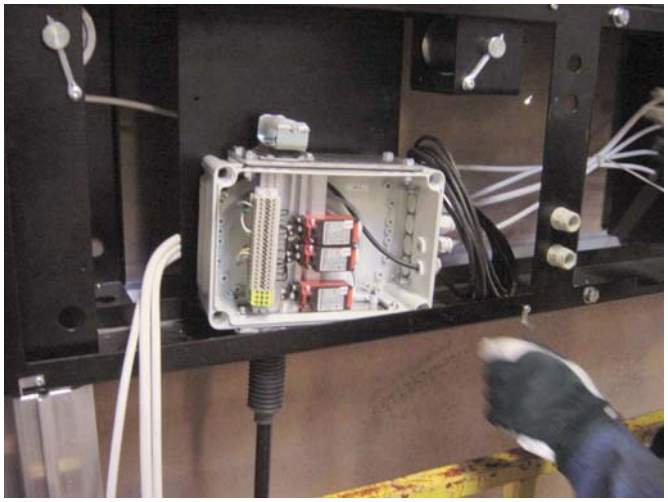




Replacement of lifting belts always require a new adjustment. If building is not fully loaded at time of installation a new adjustment/check might be necessary after building is completed.

2.12 Sensitive edge cable

Before connecting the sensitive edge cable, ensure that the power supply to the control cabinet is disconnected, so that the door cannot be operated unintentionally.

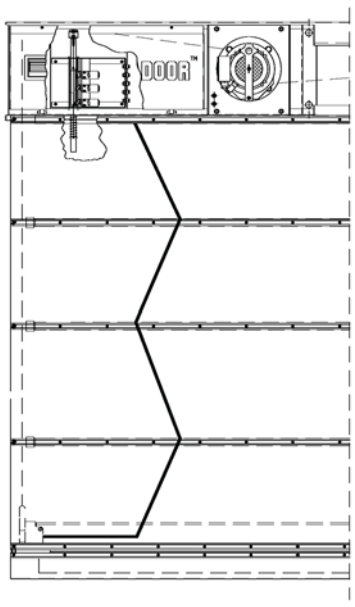




1) Loosen the safety-edge cable that lies rolled up in the header box. Thread it down through the hole in the top section.

2) Lower the door, one section at a time and pull the safety-edge cable through the intermediate sections as illustrated at point 4.

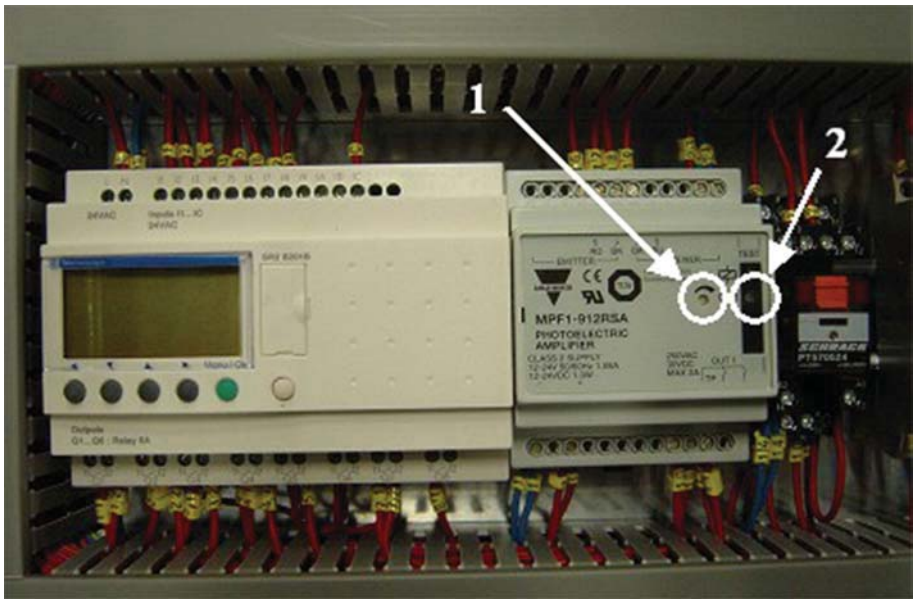




3) Connect the sensitive edge cable to the 4 terminal blocks and secure it with the clamp on the bottom section.

4) Lay the cable in a zigzag pattern.

2.13 Sensitive edge - adjustment



Open the door completely. Turn the potentiometer (1) counter-clockwise until the yellow LED (2) goes out. Then turn the potentiometer slowly in the opposite direction until the LED is lit again to somewhat reduce the sensitivity.

Perform a test operation of the door. If the closing movement is interrupted due to the sensitivity being set too high, turn the potentiometer slightly clockwise and try again.

2.14 Header box - enclosing



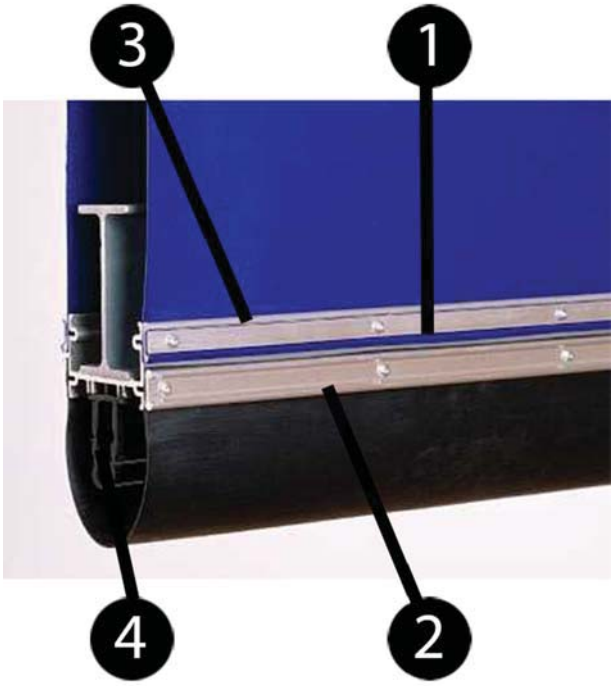




1) Install front cladding and inspection hatches. The cladding must be installed edge-to-edge (no overlapping) in the same order as at delivery, so that joints if any, are placed on the supports. Seal the joints with sealing paste.

2) Fit motor casing if any.

2.15 Fitting fabric to bottom section



1) Clamp strip alt 1
2) Clamp strip alt 2
3) Inter clamp strip alt 3
4) Rubber profile with light channel for sensitive edge

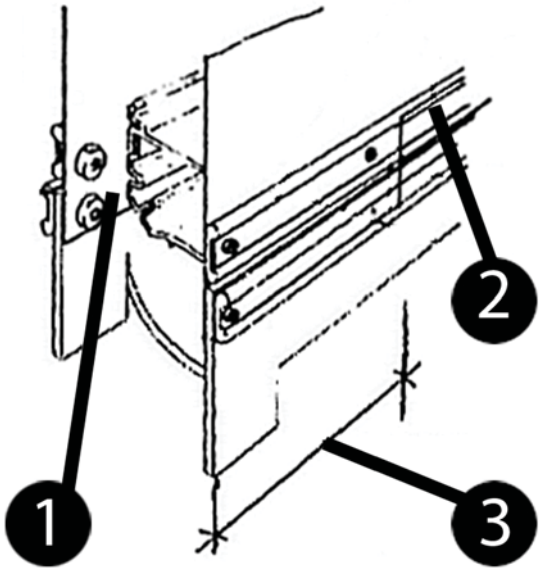
Run the door down so that the floor seal is tight against the floor. Check that the rubber profile and contact strip do not touch the floor, feeling with the hand along the whole width of the door. Distance from floor to screw in clamp strip alt 2 shall be about 9 cm.



1) Remove the upper clamp strips, alt 3 and alt 1. NB. It is important that the clamp strips do not change place when refitting them.



2) Tension the fabric towards the floor and secure the fabric with clamp strip alt 3; beginning at the middle and working outward towards the sides. Ensure that the fabric is stretched and even, especially at the edges.



1) The flap furthest out is to be folded down behind the sealing rubber and secured with screw and nylon knob against the strip and also with the first screw in the bottom section.
2) Between the ends the fabric is to be folded up and secured with clamp strip alt 1 on clamp strip alt 3.
3) Flap 20 mm wider than the distance from edge to the second screw hole.

3) Cut flaps of the fabric at each fabric edge, as illustrated above. Fold up the fabric between the flaps and fasten it with clamp strip alt 1. Loosen the two outermost screws in clamp strip alt 2. Cut the flap 20 mm below clamp strip alt 2. Thread the flap behind the sealing rubber. Secure the fabric in all clamp strips with screw and nylon knob. Refit the other screws in clamp strip alt 2.



4) Cut away surplus fabric above clamp strip.

2.16 PLC operating instructions

2.16.1 Summary

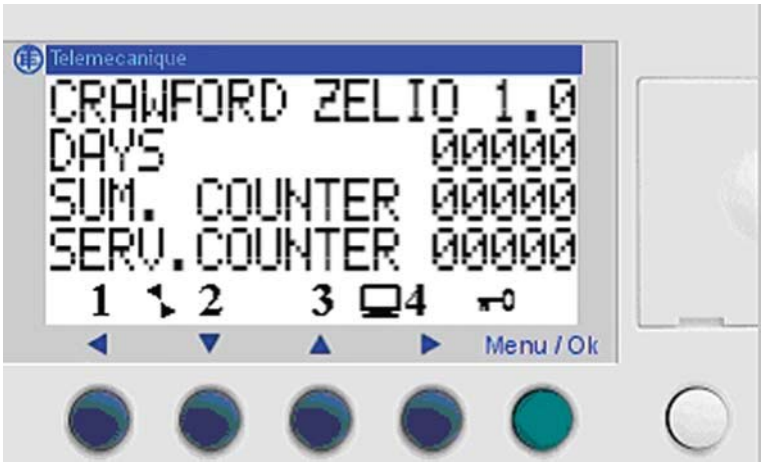
This information is valid for the 0451 control unit. The control unit can be equipped with a number of options. Optional equipment is marked with a * in the text.

The control unit contains a PLC for the setting of timers and automatic functions.

On the PLC screen the following information is displayed:

- Software version installed.
- Days of operation for the cabinet since last service.
- Number of door openings from the start.
- Number of door openings since last service occasion

If an error recognized by the PLC has occurred, an error message is displayed instead of the normal mode display. It could be as an example a defect safety edge *.



Normal mode
Note! The Menu button shall not be used! The PLC may become unusable if this button is pushed.

The PLC is programmed by Crawford before delivery, the software must not be modified.

Timers can be set for:

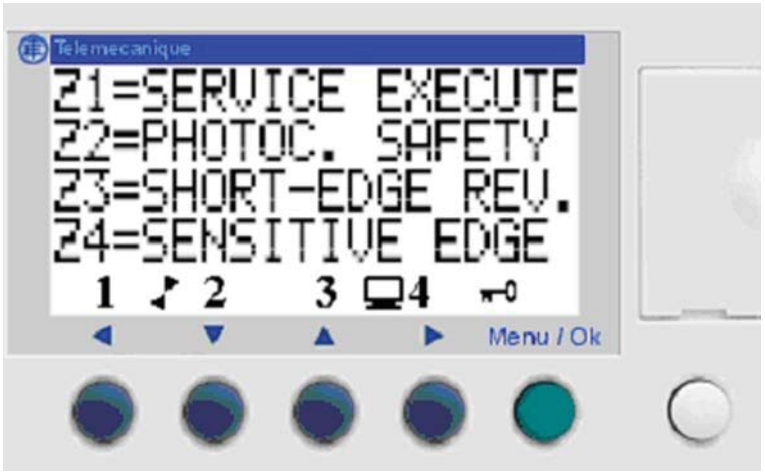
- Reduced door opening*
- Forewarning*
- Automatic closing*

2.16.2 Configuration

To configure the PLC and reset the service counter, a Configuration Menu is used. The Configuration Menu is shown for 1 minute when power is turned on. After that the display returns to Normal Mode again. Therefore, turn off the power, wait for 10 sec. and turn it on again to configure the PLC.

Push buttons Z1 to Z4: As long as one of these buttons is pushed, a help text is displayed. See FUNCTIONS.
If the door is equipped with a sensitive edge*, and it has been broken, it is possible to alter the configuration as follows:

- Turn off the power, wait for 10 sec and turn it on again.
- Press Z4 until the text “SENSITIVE EDGE NOT CHOSEN” is shown, then release the button.
- The door is now in ” hold-to-run" mode for closing and possible to operate again.



Configuration menu

When the sensitive edge is working again, the impulse control mode is activated by repeating the procedure above with the difference that “SENSITIVE EDGE CHOSEN” is shown when pressing the Z4 button.

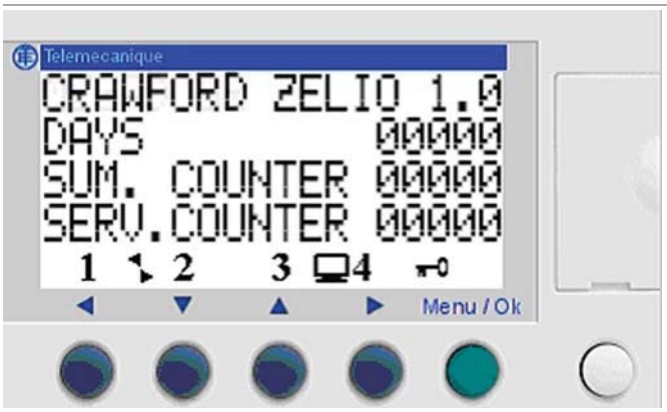
Functions:	
Z1 = Reset service	Press Z1 twice. Following text is shown: "TO RESET SERVICE COUNTER? PRESS Z1 AGAIN". Press Z1 again to execute the reset operation. The text "SERVICE EXECUTED SERVICE COUNTER IS RESET" is shown for 5 seconds.
Z2 = Photocell safety	Selected if the door is equipped with photocells or magnetic loops for safety. Press Z2 until the text "PHOTOCELL SAFETY CHOSEN" is shown. To undo the choice, press Z2 again, the text "PHOTOCELL SAFETY NOT CHOSEN" is then shown.
Z3 = Short edge reverse	If the door is equipped with sensitive edge it is possible to choose short edge reverse, which means that the door opens as little as possible when the sensitive edge is effected. Press Z3 until the text "SHORT EDGE REVERSE CHOSEN" is shown. To undo the choice, press Z3 again, the text "SHORT EDGE REVERSE NOT CHOSEN" is then shown.
Z4 = Sensitive edge	Activated if the door is equipped with sensitive edge. The door will then have impulse controlled closing. Press Z4 until the text "Sensitive edge chosen" is shown. To undo the choice, press Z4 again, the text "Sensitive edge not chosen" is then shown.

2.16.3 Time settings

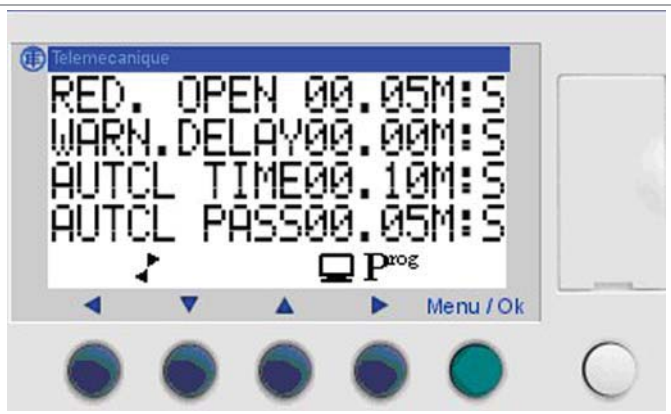
To switch between Normal Mode and Time Settings, press Z1 and Z4 simultaneously.

The timers are shown with 4 digits 00:00 (min:sec). The pre-set timers for Reduced Door Opening, Forewarning (Warning Delay), Automatic Closing and Passage Control can be adjusted in the following way:

- Press SHIFT continuously, press Z4 until the text "PROG" is shown above Z4.
- Release SHIFT and the first adjustable timer starts flashing.
- Alter between timers with Z1 or Z4 - see Functions.



Normal mode



Timers

When the setting of timers is finished, press SHIFT and Z4 simultaneously. Switch back to Normal Mode by pressing Z1 and Z4 simultaneously.

Functions:	
Z1 + Z4	Alter between Normal Mode/Timers
Shift + Z4	Start/finish setting of timers
Z1 or Z4	Alter between timers
Z2 or Z3	Change each timer (arrow up-down)

2.16.4 Error messages

There are four error messages that can be displayed on the PLC (one message at the time):
FAILURE, SAFETY CIRCUIT BROKEN
The safety circuit is interrupted. Examples:

- Emergency stop
- Normal stop button stuck or problem with contact block
- Motor circuit breaker tripped
- Main limit switch affected
- Crank switch affected

FAILURE SENSITIVE EDGE
Sensitive edge affected or damaged

FAILURE, CHECK CONTACTORS AND CONTACT BLOCKS ON CONTACTORS
The contactors do not pull although the PLC gives order to, probably caused by a failure in a contact block.

FAILURE PHOTOCELL SAFETY
Photocell or magnetic loop affected or damaged.

If problems occur with the PLC or support is needed, please contact your local Crawford representative.

2.17 Final inspection and hand over

Final inspection:
See checklist.

Marking:
The door is marked according to EN 13241-1:2003 + A1:2011.

Handing over:
Write an installation report.
Hand over and go through the following documents with the customer:
- User Manual
- Logbook

2.18 Checklist

Door No: _____ Customer: _____

Fitter: _____ Date: _____

General:

Check functioning of the door: Limit positions including main limit positions, crank switch, emergency stop and safety edge (reversing function). Ensure that the cables are neatly clamped and not damaged. The door fabric must be clean and tensioned when the door is closed. The door leaf shall run smoothly in the guide rails without any noise.

Header box		OK (x)	Incomplete due to:	Sign.
1.	Lower limit switch working			
2.	Upper limit switch working			
3.	Max. limit switch working			
4.	Motor brake operates directly on breaking the control voltage			
5.	Instruction plate, manual operation of motor			
6.	Crank available			
7.	Key in motor shaft			
8.	Lifting belt not twisted			
9.	Lifting belt does not rub			
10.	Lifting belt undamaged			
Door leaf				
11.	Fabric undamaged			
12.	No screws missing			
13.	Rubber seal on bottom profile undamaged			
14.	Rubber seal on bottom profile adjusted (cut) against floor at the ends			
15.	Sensitive edge working			
16.	Safety arresters working			
17.	Safety arresters run freely in the guide rails			
18.	Smooth guide rail joints			
Electric equipment				
19.	Stop button fitted			
20.	Cable clamping completed			
21.	Customer instructed regarding functions of the door and user manual handed over			
22.	Installation report approved and signed.			

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